**DO NOW — RECALL**

# Lesson 096 · Little Man Computer: Instructions and the A

Name: _____ Class: _____ Date: _____

**1. What is the ACC for?**

**2. What does STA X do?**

**3. What does INP do?**

**4. Half adder A1,B1: what Sum and Carry?**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Stores current/intermediate calculation.

### 2. Model answer

Stores current ACC into mailbox X.

### 3. Model answer

Reads input into ACC, replacing its former value.

### 4. Model answer

Sum0 (XOR); Carry1 (AND).

### By the end, you can…

• LMC is a teaching model of a stored-program accumulator processor: mailboxes hold instructions or data, and ACC holds the immediate working value.

• INP replaces ACC; STA copies ACC into a mailbox; LDA copies a mailbox's contents into ACC; ADD adds mailbox contents to ACC.

• OUT writes current ACC to output. Trace ACC separately from mailbox contents and PC instruction addresses.

### Vocabulary — know these terms

LMC — Little Man Computer, simplified stored-program model.

mailbox — Memory location.

ACC — Accumulator for working results.

STA — Store ACC into memory.

LDA — Load ACC from memory.

NEW LEARNING

# Read, annotate and explain

### Observe

LMC is a teaching model of a stored-program accumulator processor: mailboxes hold instructions or data, and ACC holds the immediate working value.

### Apply

INP replaces ACC; STA copies ACC into a mailbox; LDA copies a mailbox's contents into ACC; ADD adds mailbox contents to ACC.

### Evaluate

OUT writes current ACC to output. Trace ACC separately from mailbox contents and PC instruction addresses.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

### Inputs 6 then 3 to the shown program. What is FIRST, ACC after ADD, and OUT?

### Follow the reasoning

FIRST6, second INP sets ACC3, ADD FIRST gives9, OUT9.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Trace program above for inputs 4,7: ACC after each INP, STA and ADD; output. [6]

### Question 2

Write an LMC program that reads one integer, stores it as N, then loads and outputs N. [6]

**YOU DO — INDEPENDENT PRACTICE**

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Compare LDA FIRST, STA FIRST and ADD FIRST in relation to ACC and mailbox; give distinct effects. [6]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**STA FIRST overwrites ACC with the mailbox's value.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.